

Day 01: Introduction to Statistics

Foundations for Data Science

100 Days of Data Science

Why I Am Studying Statistics First

Data Science is not only about coding or building models. It is about understanding data and making correct decisions from it.

Statistics provides the logical framework to:

- Understand what data is saying
- Avoid wrong conclusions
- Build trust in results

Without statistics, Machine Learning models become black boxes and decisions become guesses.

Data vs Information

- **Data** is a collection of raw facts and numbers
- **Information** is meaningful insight extracted from data

Statistics helps convert data into information by summarizing, analyzing, and interpreting it.

What is Statistics?

Statistics is a branch of mathematics that deals with:

- Collection of data

- Organization of data
- Analysis of data
- Interpretation of data

In simple words, statistics helps us make sense of large and complex datasets.

Why Statistics is Important

Statistics is important because real-world data:

- Is large and unstructured
- Contains noise and errors
- Is often incomplete

Statistics allows us to work with uncertainty and still make reliable conclusions.

Real-World Applications of Statistics

Statistics is used in almost every field:

- **Business:** Sales forecasting, customer behavior analysis
- **Medicine:** Clinical trials and disease risk estimation
- **Government:** Census data and election surveys
- **Technology:** Data Science and Machine Learning systems

Population and Sample

Population

Population refers to the entire group that we want to study.

Examples:

- All students in a university
- All users of an application

Studying the population gives accurate results but is often impractical.

Sample

A sample is a smaller subset selected from the population.

Examples:

- Surveying 1,000 voters instead of all voters
- Testing a few products from a large batch

Key idea: A good sample should represent the population.

Why Sampling is Used

Sampling is used because:

- It reduces cost
- It saves time
- Studying full population may be impossible

Good statistical methods ensure sample results closely match population behavior.

Types of Statistics

Statistics is broadly divided into two types.

Descriptive Statistics

Descriptive statistics focuses on describing what has happened in the data.

Examples:

- Average marks of students
- Graphs like histograms and bar charts

It summarizes data but does not make predictions.

Inferential Statistics

Inferential statistics uses sample data to make predictions about a population.

Examples:

- Predicting election outcomes using surveys
- Estimating average income using sample data

Statistics in the Data Science Lifecycle

Statistics is used at every stage of Data Science:

- **Before ML:** Exploratory Data Analysis (EDA)
- **During ML:** Model evaluation and loss functions
- **After ML:** Result interpretation and validation

Statistics vs Mathematics

- Mathematics focuses on exact and ideal conditions
- Statistics deals with real-world uncertainty and variability

Statistics helps make decisions even when perfect information is not available.

Common Beginner Mistakes

- Blindly trusting averages
- Ignoring outliers
- Confusing correlation with causation

Statistics teaches us to question data instead of blindly accepting it.

Final Takeaway

Statistics is not about memorizing formulas. It is about **thinking logically using data**.

This foundation is essential for becoming a confident Data Scientist.